WEEK 7 OF 15 · STAGE 2: GCAM IN DEPTH

Inside GCAM

How a big model balances the world's markets

New R skill rgcam: connect to the output database, add scenarios, run queries, save a project file

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	**Week 7 Inside GCAM**	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Describe GCAM's regions, time steps and the markets it balances
2. Explain how GCAM finds prices one period at a time
3. Calculate logit shares for three technologies by hand
4. Run the Reference scenario and pull its results into R with rgcam

? **Big question** How can one model decide, for 32 regions at once, how much coal, gas and solar gets built?

One auction, thousands of markets

In week 4 you found one price where supply met demand. Now imagine doing that for every fuel, crop and region in the world, all at once.



If coal is cheapest, should it get the whole market?

AN EVERYDAY PICTURE

GCAM is a factory with an auction house inside

Picture a huge factory with 32 screens on the wall, one per region. An auctioneer shouts prices for oil, electricity, wheat and water. He keeps adjusting until every screen turns green, meaning supply equals demand everywhere. Then his stopwatch beeps and he moves 5 years ahead.

What it means for models

1. Screens = regions; prices = markets
2. Green screens = equilibrium in every market
3. Stopwatch = 5-year time steps

NUMBERS TO REMEMBER

GCAM's world in three numbers

32

energy-economy regions, such as Japan, USA and EU-15

5

years between solved periods

235

river basins where water markets clear

Land is split into more than 300 smaller subregions. Source: GCAM v8.2 documentation (Common Assumptions, Overview).

HOW IT WORKS

How GCAM solves one period

1

Guess prices

Start from a price for every market

2

Add up

Work out supply and demand at those prices

3

Find the gaps

Demand minus supply in each market

4

Adjust

Raise prices where short, lower where in surplus

5

Next period

When all gaps are near zero, step 5 years on

The same loop you wrote with uniroot in week 4, for every market at once

COMPARE

Two ways a model can handle time

GCAM: one step at a time

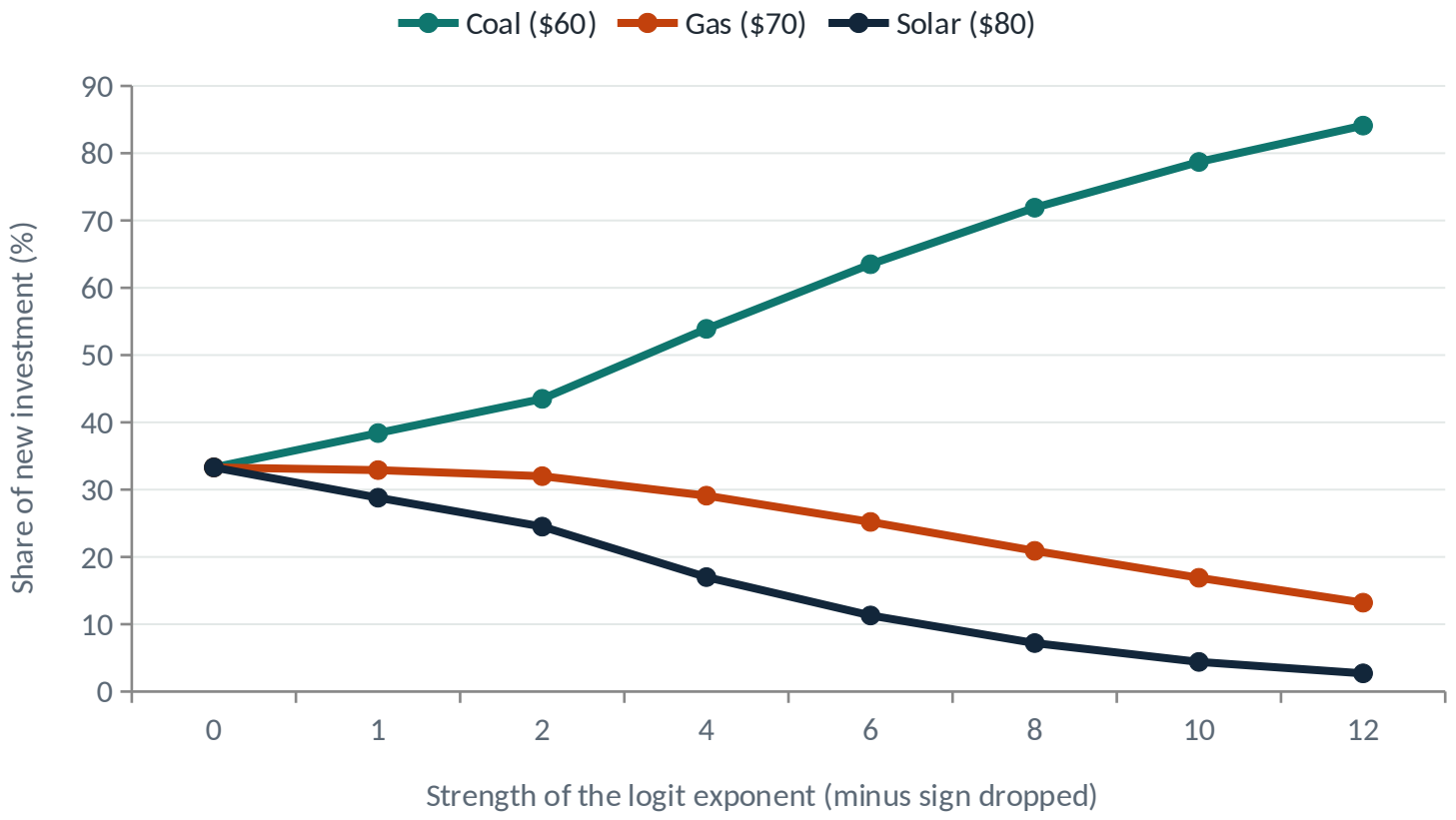
- Solves 2025, then 2030, then 2035
- Decisions use today's prices only
- Plants, land and resources carry over

DICE (week 5): sees the whole future

- Chooses the whole path at once
- Knows future damages and costs
- Called perfect foresight

PICTURE THE NUMBERS

Cheaper wins more, but never everything



Week 4 lab costs, equal share weights. Illustrative, computed from the logit formula.

Read it like this

1. At 0, cost does not matter: equal thirds
2. At 4, the split is 54 / 29 / 17
3. The bigger the exponent, the closer to winner-takes-all

Worked example: logit shares by hand

Problem Coal $60, gas $70, solar $80 per MWh. Equal share weights. Exponent −4. Share = cost^−4 ÷ sum of all cost^−4.

1. Coal: 60^−4 = 1 ÷ 12,960,000 = 7.72 × 10^−8
2. Gas: 70^−4 = 1 ÷ 24,010,000 = 4.16 × 10^−8
3. Solar: 80^−4 = 1 ÷ 40,960,000 = 2.44 × 10^−8
4. Sum = 7.72 + 4.16 + 2.44 = 14.32 (× 10^−8)
5. Shares: 7.72 ÷ 14.32 = 54%, 4.16 ÷ 14.32 = 29%, 2.44 ÷ 14.32 = 17%

Answer **Coal 54%, gas 29%, solar 17% of new investment**

Quick check

Two technologies cost $50 and $100 per MWh. Equal share weights, exponent −2. What share does the $50 one get?

- **A** 50%
- **B** 67%
- **C** 80%
- **D** 100%

Quick check: answer

Two technologies cost $50 and $100 per MWh. Equal share weights, exponent −2. What share does the $50 one get?

A 50%

B 67%

C **80%** ✓

D 100%

HOW IT WORKS

Files in, database out

1

Input files

XML files built in R by gcamdata

2

Configuration

configuration.xml lists the inputs; later files win

3

GCAM core

Reads inputs, solves each period, writes a log

4

Database

Results stored in database_basexdb

5

R

rgcam runs queries and returns tables

Rule for every lab: never edit an original input file. Copy it and add the copy last.

KEY IDEA

The four scenarios you will meet in this stage

1

Reference

GCAM's own baseline with no new climate policy. Every other scenario is read as a difference from it.

2

CheapSolar (week 8)

Same as Reference, but solar power is assumed cheaper in the future.

3

CarbonPrice (week 10)

A global price on CO2 that starts in 2025 and rises 5% a year.

4

NetZero (week 10)

A limit on emissions that falls to zero by 2050. GCAM finds the price.

WATCH OUT

Common mix-ups

Not quite	Better
Dynamic-recursive means GCAM ignores time	Plants, land and resources carry over between periods. It only means no decision sees future prices.
A logit share is the chance that a technology wins	It is the fraction of new investment each option gets in that period.
GCAM predicts the future	It works out what follows from its input assumptions, consistently across all markets.

NEW WORDS THIS WEEK

Words to add to the class glossary

GCAM The Global Change Analysis Model, built at PNNL's Joint Global Change Research Institute. It links energy, economy, land, water and climate.

Time step The jump between model years. GCAM solves one year every 5 years, for example 2025, 2030, 2035.

Dynamic-recursive Solving one period after another, without knowing future prices. What was built earlier carries over to the next period.

Logit exponent A number that sets how strongly market shares react to a cost gap. Bigger (more negative) means the cheapest option grabs more.

Share weight A tuning number for each option, set so the model reproduces past data. It also lets new technologies phase in.

Lab 7: run Reference and query it from R

Start your own Reference run on the course server, then connect to the instructor's database, pull results into R and draw your first GCAM charts.

Script: labs/lab7.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Start a Reference run (Terminal tab)

These are shell commands, not R. Type them in RStudio's Terminal tab. The run keeps going after the lab ends.

```
cd ~/gcam/exe
cp configuration_ref.xml configuration.xml
nohup ./gcam.exe -C configuration.xml &
tail -n 3 logs/main_log.txt
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Connect to the database

localDBConn points R at the results. addScenario runs every query in the query file for one scenario and saves a project file.

```
library(tidyverse)
library(rgcam)
db_dir <- "/course/gcam/output"  # from instructor
conn <- localDBConn(db_dir, "database_basexdb")
listScenariosInDB(conn)

dir.create("output", showWarnings = FALSE)
prj <- addScenario(conn, "output/gcam_course.dat",
  "Reference", "gcam-files/course_queries.xml")
listScenarios(prj)
listQueries(prj)
```

Get one query and plot it

Every query comes back as a long table, so your dplyr and ggplot2 skills work straight away.

```
co2 <- getQuery(prj, "CO2 emissions by region")
glimpse(co2)
japan <- filter(co2, region == "Japan")  # your region
ggplot(japan, aes(year, value)) +
  geom_line() +
  labs(y = japan$Units[1])
```

Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Electricity by fuel, then save

A subsector is a fuel group such as coal, gas or solar. Save the project so next time you can just load it.

```
elec <- getQuery(prj, "elec gen by subsector") |>
  filter(region == "Japan")
ggplot(elec, aes(year, value, fill = subsector)) +
  geom_col()
saveProject(prj)
# next time:
# prj <- loadProject("output/gcam_course.dat")
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

LAB · CHECK YOUR WORK

Did it work?

You should see

- listScenariosInDB shows the four course scenarios
- Your CO2 chart shows one smooth line for your region
- The electricity chart stacks coal, gas, solar and other fuels by year

Worksheet questions

1. Which fuel groups supply most electricity in your region in 2020 and in 2050?
2. Convert your region's 2050 CO2 emissions from MtC to MtCO2.
3. Using the Kaya factors from week 3, suggest what drives your region's emissions path.

RECAP

Take-home messages

1. GCAM balances every market in every region, one 5-year period at a time
2. Dynamic-recursive: no foresight, but what is built carries over
3. The logit rule shares new investment by relative cost: cheaper wins more, never all
4. Results live in a database; rgcam turns queries into R tables

Next: week 8

Week 8 opens the energy system: how technologies compete on cost, and what happens when you make solar cheaper.

Practise

Interactive course, week 7: try the logit explorer and the 5-question quiz.

Readings: Calvin et al. (2019), GCAM v5.1, Geoscientific Model Development, sections 1 and 2; rgcam package guide; GCAM v8.2 documentation: Overview, Common Assumptions, Economic Choice, Solution and Computation, How to Get Started